MERIDIAN ♥

Memory-**E**ncoded **R**ituals for **I**ntimacy across **D**istance in an **I**nteractive, **A**synchronous **N**eighbourhood

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(a) The whole town, drawn in code (b) The square: guide, partner, cats (c) Prologue: the birthday cake (d) Finale: the wedding

Figure 1: Yi Town at a glance. (a) The town map: a 24×20 grid of 16-pixel tiles with nine numbered houses, one per level, painted entirely by code. (b) The fountain square in play, top after every level is cleared (gold flags, a cat sunbathing by the pond) and bottom halfway through; the player meets the partner, the guide Rayford and the cats. (c) The birthday prologue: nine candles in the dark at Shenzhen midnight, and ten strawberries set in a heart. (d) The wedding in the fountain garden, officiated by Rayford with the two cats as guests.

Abstract

Two people met on 20 September 2026 and have since lived thirteen hours and one Pacific Ocean apart, one in Shenzhen, China, the other in College Station, Texas. This paper reports the design and implementation of Yi Town, a bilingual, browser-only game made by this couple for themselves, first as a birthday gift and then as a shared artefact that keeps growing. Each player walks a pixel avatar through a small town whose nine houses are levels, and each level turns one shared memory into a game mechanic: a vow spoken aloud, street views to name on a map of six places, a tattoo handwritten on a drawn arm, a timeline to put back in order, Gomoku endgames whose answers trace a heart, songs named from a passage, the stars of two birth signs joined by hand, missing characters restored to classical Chinese love poems, and a procedurally modelled ring tried on through hand tracking. A birthday prologue and a wedding finale play as pixel cutscenes. Using research through design and autobiographical design, we state six design commitments (memory as mechanics, earned access, intimacy without a server, parity between the two languages, art drawn in code, and secrecy as narrative tension) and show how the system realises each of them. Progress is modelled as a grow-only set, so merging two devices' progress needs no server. It travels between phones as a code of at most four characters guarded by an ISO 7064 MOD 37,36 check character. The sky follows each player's own clock, and a unit test keeps the final surprise out of every string shown before it. The system comprises about 8,400 lines of strict TypeScript and 153 unit tests and was built in 109 commits over five days; its source is public at https://github.com/rayford295/yi-town. We reflect on what designing for an audience of exactly one other person reveals about privacy, asynchrony and ceremony in intimate computing.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Collaborative and social computing theory, concepts and paradigms*; • **Applied computing** → *Computer games.*

Keywords

intimate computing, long-distance relationships, autobiographical design, research through design, games for two, local-first software, bilingual interfaces, pixel art, ceremony, surprise

1 Introduction

Jingyi and Yifan met on 20 September 2026. From the first day they have lived on opposite sides of the Pacific: Jingyi in Shenzhen, Guangdong, where she works in product management, and Yifan in College Station, Texas, where he is a doctoral student. Between their clocks lie thirteen hours, so that one of them is always walking

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home in the dark while the other is starting a morning. Their next meeting is fixed by a plane ticket Yifan drew by hand on their first video call (flight *CYF925*, Shenzhen to New York, 5 February 2027, gate *LOVE910*, seat *99A*).

Long-distance couples are not rare, and their communication can be more intimate, not less, than that of couples who live together [15, 32]. Human-computer interaction (HCI) has long designed for such couples, from devices that pass a squeeze or a hug across a distance [5, 24, 33] to systems that share the empty moments of a day [20, 34] or one bit of "I am thinking of you" [17]. These systems are built by researchers for couples whom the researchers then study. The artefact in this paper reverses that relation. Yi Town was built by one member of a couple, for the other and for himself, out of material that only the two of them hold: the date they met, the songs they share, the photos, a first game played on a Zoom whiteboard, the stars of their two birth signs.

Yi Town is a web game for exactly two players (Figure 1). After a question that only the two can answer, a player picks who they are and walks a twelve-by-sixteen-pixel version of themselves through a small town. The partner waits by the fountain. Each of the nine houses is a level made from something that happened to the couple; a cleared house flies a gold flag, and the two cats of the household, Coffee and Duoduo, sit at the next locked doors. The first time a player arrives, a cutscene shows Yifan baking a birthday cake against the Shenzhen-midnight clock. When both players have reached the last house, a letter opens, and after it a wedding.

We call the design Meridian, after the pair of rings that waits in the last house. A meridian is also the line by which the world keeps its time: thirteen hours of them lie between the two players, and the town is our attempt to live on one meridian together. The acronym states the argument of the paper: shared memories, encoded as small rituals, can carry intimacy across a distance when they are placed in a neighbourhood that both people can enter, but never at the same time.

This paper makes four contributions.

(1) A **design case** of an intimate game built autobiographically, for an audience of one other person, documented through its design log of 109 commits and 28 versions (Sections 3 and 4).
(2) Six **design commitments** for intimate play across distance, each traced to concrete mechanisms (Table 2).
(3) A **serverless architecture** for two-person progress: progress as a grow-only set whose merge is a union, carried between phones as a short checksummed code, with a sky that follows each player's local time and a test that guards the final surprise (Section 5).
(4) A **reflection** on privacy, asynchrony, bilingualism and ceremony when the designer, the player and the subject of the design are the same two people (Section 7).

2 Related Work

2.1 Mediating Intimacy at a Distance

Vetere et al. studied how couples and families express intimacy and argued that technologies for strong ties should support the expressive, the physical and the playful, not only the informational [35]. Hassenzahl et al. reviewed 143 intimacy-mediating artefacts and grouped them into six strategies: awareness, expressivity, physicalness, gift giving, joint action and memories [11]. Many such systems are deliberately minimal channels, such as the Virtual Intimate Object, a single button whose meaning the couple negotiates [17]. Neustaedter and Greenberg found that some long-distance couples leave a video call running for hours to share "being together" rather than to talk [26]; CoupleVIBE conveyed a partner's location by vibration [3]; SyncDecor linked everyday appliances in two homes [34]. Yi Town sits in Hassenzahl's *memories* and *gift giving* strategies, but differs in scale and authorship: it is not a minimal channel but a whole world, and its content is the couple's own history rather than a live signal.

2.2 Autobiographical and First-Person Design

Neustaedter and Sengers defined autobiographical design as design research drawing on genuine use by the people who built the system [27], and Desjardins and Ball documented its tensions, including the blurred line between designer and participant, and the intimacy of the data involved [6]. First-person methods, "a sample of one" [21], accept that what is learned is situated and deep rather than general. Research through design treats the artefact itself as a carrier of knowledge [7, 38]. Yi Town is autobiographical twice over: it is built by one of its two users, and it is built *out of* the two users' lives.

2.3 Play, Ritual and Games for Two

Huizinga described play as a "magic circle" set apart from ordinary life [12]; Salen and Zimmerman made the circle a design tool [29]; Juul argued that games are half real rules and half fiction [16]. Sicart argued that play appropriates the world and its objects [31], and Isbister argued that games build connection between players through shared action [14]. Ludic design in HCI values curiosity and ambiguity over task efficiency [8, 9], and slow technology designs for reflection rather than throughput [10, 28]. Aron et al.'s procedure for generating closeness through escalating mutual disclosure [2] is, in a sense, a level structure. Yi Town takes the game form literally: locks, levels, a guide, a prologue and a finale, and play that follows the order of a relationship that is still being lived.

2.4 Local-First and Serverless Software

Kleppmann et al. argued for local-first software that keeps the primary copy of data on the user's device and works offline [18]. Conflict-free replicated data types (CRDTs) guarantee convergence without coordination; the simplest, the grow-only set, merges by union [30]. In intimate computing, keeping data on the device is as much a matter of feeling as of security: photos, voice notes and letters between two people should not live on a third party's server. Personal images are also more revealing than they look, since a large multimodal model can infer where a photograph was taken from its visual content alone [36]. Yi Town applies the grow-only set to a two-player game and replaces the network with a code that can be read aloud.

Table 1: Key dates encoded in the system (`src/core/config.ts`).

Date	Moment
2026-09-20	We met; day 1 of every counter in the town
2026-09-25	Yifan's birthday (Libra)
2026-10-17	First video call; the hand-drawn ticket CYF925
2026-10-23	First Gomoku game, on a Zoom whiteboard
2026-10-24	First commit of Yi Town
2026-10-31	Jingyi's birthday (Scorpio); first release
2027-02-05	CYF925, Shenzhen → New York; the countdown's target

3 Design Context and Method

3.1 The Two of Us

Table 1 lists the dates that the system encodes. The couple speak Mandarin to each other; Yifan works in English, Jingyi in Chinese and English. Both use phones as their main devices, and Jingyi's network in mainland China blocks some services (YouTube, Google Fonts) that Yifan's does not. These facts became requirements: every string exists in both languages; no feature may depend on a service blocked on either side; and everything must run on a phone.

3.2 Method

We follow research through design [7, 38] in its autobiographical form [6, 27]. Yifan acted as designer, author of the content and (with an AI coding assistant; see the Declaration of Generative AI Use) implementer; he also playtested every level as both players. Jingyi is the intended first player; her birthday, 31 October 2026, is the release date. The design log consists of the repository history (109 commits between 24 and 28 October 2026), a changelog of 28 versions (0.1.0 to 1.17.0), and the design specifications written before each major step (the site, the game of levels, the town). We treat the corrections the owner made to his own system as data: each one records a moment where a plausible default was wrong for *these* two people.

3.3 Six Design Commitments

Six commitments emerged and hardened over the five days. Table 2 maps each to the mechanisms that realise it.

D1 Memory as mechanics. Every level's rule is derived from something that happened. The first game was Gomoku on a Zoom whiteboard, so a level is Gomoku. The ticket was drawn on the first call, so the first level counts down to that flight.

D2 Earned access. Levels unlock in order, and only through their doors: there are no menu shortcuts. The progression mirrors the relationship, from the day they met to what comes next.

D3 Intimacy without a server. Nothing leaves the device: no accounts, no analytics, no database. Progress crosses phones as a code the two can send in a chat.

D4 Bilingual parity. Chinese and English are equal: neither is a translation of the other, and a test fails if a string is missing in either.

Table 2: Design commitments and the mechanisms that realise them.

Commitment	Realised by
D1 Memory as mechanics	Nine levels, each from one event (Table 3); two "days" mapped from real routines
D2 Earned access	Ordered level registry; router guard; door-only entry; cats guarding the next locks
D3 No server	`localStorage` per player; grow-only set; progress codes; PWA offline cache
D4 Bilingual parity	One 791-line dictionary with both languages per key; tests on both
D5 Drawn in code	Sprites and tiles as strings; banner and icons generated from them
D6 Secrecy	`secret.test.ts` over the whole dictionary; a guide whose lines never name the ring

D5 Art drawn in code. Every tile and sprite is a string of characters in the source; the game ships no image files of its own, so there is no licensed art and the art can be changed as easily as text.

D6 Secrecy as narrative tension. The last house holds a surprise, which nothing before it may reveal; a unit test enforces this across every user-visible string.

4 The Experience

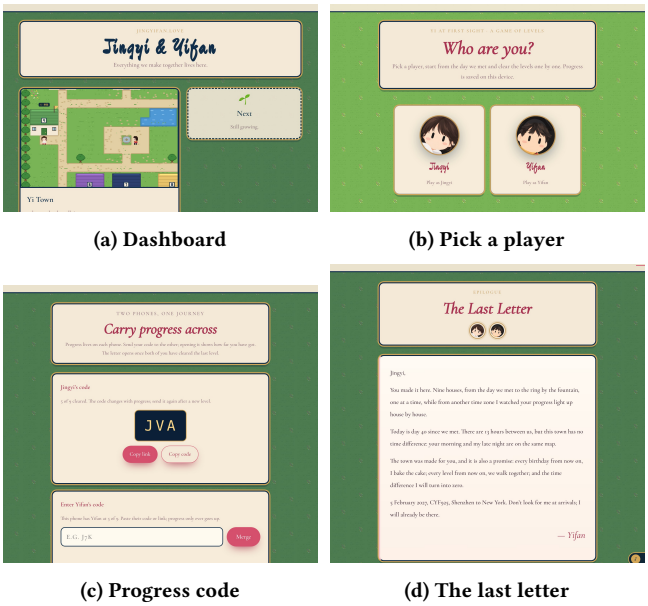
4.1 The Front Door

The site at `jingyifan.love` opens on a single question. The answer, a date both players know, is accepted in several spellings (`9.20`, `0920`, `9月20日`, `20260920`); it is reduced to its digits and compared by SHA-256 hash [25], so the answer is not written in the bundle. Behind it is a dashboard that lists everything the couple build under the domain (Figure 2a), of which Yi Town is the first. Choosing the town asks *Who are you?* (Figure 2b). Progress is kept per player on each device, so both can play on one phone.

4.2 The Town

The town (Figure 1a) is a 24×20 grid of 16×16-pixel tiles. Paths connect nine houses around a fountain square and a pond. Each roof has its own colour and its level number, and a cleared house flies a gold flag. The player moves with the arrow keys, by tapping a tile (the avatar walks a shortest path to it), or with an on-screen d-pad. Walking into a door opens a talk box with the level's one-line story, for example "Our first game was on a Zoom whiteboard. Now you place the last stone."

Four kinds of character live in the town (Figure 3). The *partner* waits by the fountain and reports how far they have got ("I am waiting here. I have cleared 5 levels. You?"). *Rayford*, Yifan in a suit, is a scripted guide south of the fountain: he explains the project, lists the levels, hints at the next one and walks the player to its door. Two Ragdoll cats, *Coffee* (咖啡) and *Duoduo* (哆哆), guard the locks: Coffee sits at the next locked door and Duoduo at the one after. When a door opens they walk to their new posts, and with nothing left to guard they sunbathe by the pond. Bumping into a cat shows a photo of the real cat. Two signposts lead out of the



(a) Dashboard (b) Pick a player

(c) Progress code (d) The last letter

Figure 2: Around the town. (a) The dashboard lists every product under the domain. (b) Each device keeps one progress per player. (c) Progress crosses phones as a code (here JVA). (d) The epilogue opens once both players have cleared the last level.



Figure 3: Every character in Yi Town is a string of characters in the source. From left: Jingyi, Yifan, Rayford, Yifan as the baker, Jingyi crowned, the bride and the groom (four facings each), then the cats Coffee and Duoduo. Bodies are 12×12 pixels over four rows of shared legs.

square to the two cities (Section 4.4), and a red mailbox holds voice notes the partner has left.

The sky follows the clock in the player's own city: Jingyi walks the town in Shenzhen time and Yifan in College Station time, so at the same instant one sees noon and the other sees lit windows under a night wash. The whole site stands on the same pixel meadow, and every heading is a cream sign in the town's palette, so leaving a house never leaves the world.

4.3 Nine Levels

Table 3 lists the levels with their source memory, mechanic and input modality, and Figure 4 shows them in play.

1. The Day We Met. The first level shows the day count since 20 September, the time in both cities with day and night icons, and a countdown to the flight. To clear it, each player reads a vow, rewritten from the wedding formula as the vow of "becoming each other's partner once more", and says one line aloud: 我爱你，陈静仪，我愿意 ("I love you, Jingyi Chen, I do"), with the partner's full name. Browser speech recognition [19] hears Mandarin names inconsistently, so the matcher removes everything except Han characters and Latin letters and accepts any of eight or nine homophone spellings of each name (for example 陈静怡, 陈静宜). A typed fallback keeps the level playable in browsers that cannot listen.

2. Love Map. Six places matter to the couple: Texas, Los Angeles, Japan, Shenzhen, Hong Kong and New York. The street views are public photographs of well-known sites from Wikimedia Commons rather than the couple's own pictures, since a personal photograph can give away far more of a place than its owner intends [36]. Before each heart appears on a Leaflet map, the player names a street-level photograph from four options (Figure 4b), and a wrong pick reshuffles the options. Great-circle arcs connect the two home cities.

3. Tattoo. The player handwrites a line or a small drawing on an arm or a shoulder, drawn in SVG, and inks it. The tattoo is kept on the device and can be saved as an image to send.

4. Moments. The timeline holds eight dated memories, each with photos and a story in both languages, from hearts made from both sides of the ocean (day 21) to a heart drawn on the Gomoku board (day 35). The player orders five of them by tapping pairs to swap, then reassembles a 3×3 portrait of the partner.

5. Gomoku. Each of ten positions on a 15×15 board has exactly one winning point for the player's own colour (purple for Jingyi, green for Yifan, as in their first game). The ten answers are the ten points of a heart; solved answers stay on the board in gold, so the heart draws itself as the player progresses. Positions are generated from three patterns (a four closed at one end, a broken four, three-and-one) placed around each heart point in varied directions with decoy stones, and a test checks that each has a unique winning move. Gomoku is a first-player win under optimal play [1]; win-in-one puzzles avoid that question entirely and make every move feel like a find.

6. Our Songs. Six songs the couple share (among them 只对你有感觉 and 天下第一好) play a passage through the official embedded players: QQ Music for the player in China and YouTube for the player in the United States, each switchable. The site hosts no audio. The options unlock only after the passage has been heard.

7. Libra and Scorpius. Yifan is a Libra (25 September) and Jingyi a Scorpio (31 October); in the sky the two constellations are neighbours, and Libra's two brightest stars still bear the Arabic names of the Scorpion's claws, *Zubenelgenubi* and *Zubeneschamali*. The level projects 22 stars from J2000 coordinates and magnitudes [23] and asks the player to draw both figures, 20 edges in all.

8. Us, in Old Verse. Five classical lines about love lose their second half, to be restored from a bank of characters: Su Wu's farewell to his wife, 结发为夫妻，恩爱两不疑; Guan Daosheng's *Song of You and Me*; a couplet of Li Shangyin; the paired birds of the sixth-century anthology *New Songs from a Jade Terrace* [4]; and an anonymous couplet. Each solved line shows a plain English gloss.

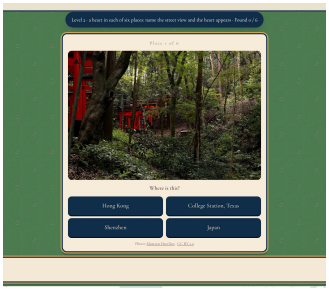
9. Meridian. The last level is a pair of rings, called *Meridian*, modelled procedurally in Blender from a Python script (Figure 5). The pair share a signature: a yellow-gold line inlaid around a platinum

Table 3: The nine levels: each is a memory turned into a rule (D1). Modalities: *V* voice, *T* touch or pointer, *C* camera, *K* keyboard alternative available.

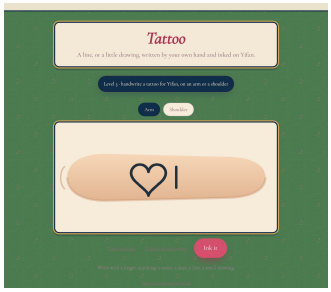
#	Level	Source memory	Task (clear condition)	Input	Data
1	The Day We Met	20 September 2026; the vow	Answer a question, then say 我爱你，*<name>*，我愿意 aloud; homophones of the name accepted	V/K	1 vow per player
2	Love Map	Where each studied; where they will meet	Name six street views; each right answer drops a heart on the map to collect	T	6 places
3	Tattoo	A promise to write on each other	Handwrite a line on an arm or shoulder, then "ink it"	T	strokes
4	Moments	The first 40 days	Order five memories, then reassemble a 3 × 3 portrait of the partner	T	8 memories
5	Gomoku	First game on a Zoom whiteboard	Ten win-in-one endgames; the ten answers form a heart	T	10 positions
6	Our Songs	The six songs they share	Hear a passage, pick the title from four	T	6 songs
7	Libra and Scorpius	Their birth signs, neighbours in the sky	Join 22 stars along 20 edges	T/K	J2000 positions
8	Us, in Old Verse	Classical love poetry	Restore the missing half-line of five poems character by character	T	5 poems
9	Meridian	What comes next	Try the ring on through the camera and take a photo	C	3 GLB models



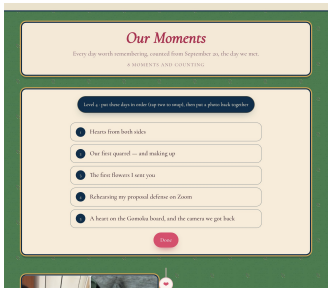
(a) 1 The Day We Met



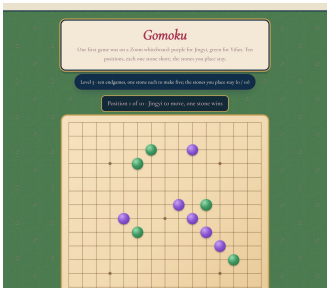
(b) 2 Love Map



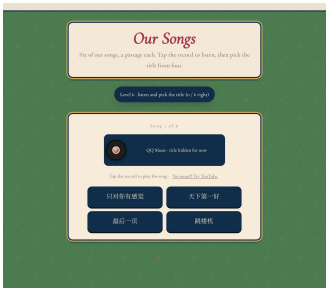
(c) 3 Tattoo



(d) 4 Moments



(e) 5 Gomoku



(f) 6 Our Songs



(g) 7 Libra and Scorpius



(h) 8 Us, in Old Verse

Figure 4: Eight of the nine levels in play (the ninth is shown in Figure 5). (a) Day counter, twin clocks and the countdown to CYF925. (b) Name the place in the street view. (c) A heart inked on an arm. (d) Put the first days in order. (e) Position 1 of 10: one stone wins. (f) Name the song. (g) Libra's "claws" and Scorpius, before they are joined. (h) Su Wu's line 结发为夫妻，恩爱两不疑 with its second half to restore.

band like a meridian on a globe, a hidden sapphire inside the bottom, and an inner engraving that carries the *other* person's name and birthday, the date they met and the coordinates of the airport where they will meet (40.64°N 73.78°W). Her ring exists in two designs, a solitaire on four compass prongs (v1) and the current flush-set stone sunk into his band's profile (v2); the page keeps both as a design history. The level is cleared only one way: the player tries the ring on through the camera and takes a photo (Section 5.5).

4.4 Two Days

Two signposts lead from the square to a day in each city (Figure 6). Each is a second map with the person's stops: Yifan's home, gym, his advisor's office, the golf course and the pool in College Station; Jingyi's home with the cats at the door, her drive, the office and Shenzhen Bay. Each stop has hours, and the person's sprite stands wherever the clock in their own city puts them. A partner who opens Yifan's day at 9:49 p.m. in Shenzhen finds him at 8:49 a.m. at his advisor's office. The schedules began as guesses and were

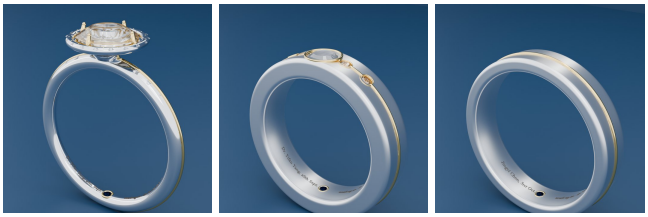


Figure 5: The Meridian pair, path-traced from the procedural Blender model: hers v1 (solitaire, halo and graduated shoulder stones), hers v2 (flush-set 4.2 mm stone in a 5 mm flat band), and his (brushed flat comfort-fit band). The gold meridian runs around each band.



(a) A day of Yifan's **(b) A day of Jingyi's**

Figure 6: The two cities as maps. The person stands at the stop whose hours contain the current local time in their own city: (a) Yifan at his advisor's office at 8:49 a.m. in College Station; (b) Jingyi at home at 9:49 p.m. in Shenzhen, the same instant.

corrected by the owner: Yifan drives rather than taking the bus; Jingyi drives a Tesla rather than taking the metro; and the Shenzhen Bay stop keeps a joke between them about where they will one day live.

4.5 Prologue and Finale

Two cutscenes frame the game (Figure 7). They run in a small pixel theatre: a 128×80 canvas scaled without smoothing, actors built from the town's sprite sheets, a darkness layer with candle-like glows, particles, a synthesised score, and a caption box for words and buttons.

The birthday prologue plays once per player before the town opens. In College Station it is 9:47 a.m.; in Shenzhen 10:47 p.m., 73 minutes before Jingyi's birthday. Yifan, in a chef's hat, bakes the cake against the Shenzhen clock. The player helps by whisking the batter (twelve taps), placing ten strawberries in the same heart that the Gomoku answers make, and watching nine candles lit (nine levels; in Chinese 九 "nine" sounds like 久 "for a long time"). At midnight in Shenzhen the lights go out, the door opens on the town ("In Yi Town there is no thirteen-hour time difference"), everyone sings, the player makes a wish and blows the candles out, by breathing into the microphone or holding a button.

The wedding opens after the last letter, once both players have cleared the last level. In the fountain garden at dusk, Rayford reveals why he had worn a suit all along: he is the master of ceremonies. The groom waits under the arch, the bride walks the aisle to Mendelssohn's *Wedding March*, both say "I do" (aloud if the browser can listen; "I don't" is not accepted), Coffee carries the rings down the aisle on a red pillow, the Meridian rings turn in 3D, and the scene ends in a kiss, bells, doves and a certificate to keep (Figure 8). Afterwards the town dresses the two sprites as bride and groom.

5 System

5.1 Architecture

Yi Town is a static single-page application in strict TypeScript, built with Vite and served from GitHub Pages on the couple's own domain (Figure 9). There is no backend. Each feature is a folder (`src/features/<name>/`) with its page module, styles, data, tests and a README, and each page is a function `render(root)` $\Rightarrow$ `cleanup` loaded lazily by a path router, so the main bundle carries only the shell and the town. Heavy modules (Leaflet for the map, three.js for the ring, MediaPipe for the try-on) load only inside the level that needs them. A service worker precaches the shell, so the town installs as an app and works offline; media is cached on first use and refreshed in the background.

5.2 The Quest Model

Let $L = (\ell_1, \ldots, \ell_9)$ be the ordered level registry and $C_p \subseteq L$ the set of levels player p has cleared on a device. A level is unlocked iff it is first or its predecessor is cleared:

$$\text{unlocked}_p(\ell_i) \iff i = 1 \ \vee \ \ell_{i-1} \in C_p. \tag{1}$$

Clearing adds to C_p and nothing ever removes from it, except an explicit reset. The router enforces the lock, and a second guard requires that a level be entered through its door: walking onto the door sets a one-shot token in `sessionStorage`, and a level route without the token redirects to the town. Level numbers are derived from the registry, so inserting a level (the Poems level was inserted before the last one during development) renumbers every house, sign and string at once; a test fails until the new house has a door that the town's path search can reach.

5.3 Progress without a Server

Each phone holds its own C_p. Because C_p only grows, it is a grow-only set (G-Set), the simplest state-based CRDT [30]: the merge of two replicas is their union,

$$C_p \sqcup C'_p = C_p \cup C'_p, \tag{2}$$

which is commutative, associative and idempotent. Two phones that exchange codes in any order, any number of times, therefore converge, and importing an old code can never take progress away. The only channel needed is one the couple already use every day: a chat message.

Progress codes. A code encodes one player and a bitmask of cleared levels, $m = \sum_{\ell_i \in C_p} 2^{i-1}$, written in base 36 after a player letter (`J` or `Y`) and followed by one check character. With nine levels $m \le 511 = \mathrm{E7}_{36}$, so a code is at most four characters: `JVA` says that Jingyi has cleared $m = 31$, the first five levels (Table 4). The check



Figure 7: The two cutscenes, left to right. Top, the birthday prologue: the kitchen at 9:47 a.m. in College Station (10:47 p.m. in Shenzhen) with both clocks on the wall; whisking the batter; nine candles on the finished cake; singing at Shenzhen midnight with the cats. Bottom, the wedding: the guests take their seats; the bride walks the aisle; Coffee brings the rings; the Meridian rings in 3D over the ceremony.



Figure 8: The certificate at the end of the wedding, saved as an image through the phone's share sheet. It records the day of the ceremony, the day they met and the day they will meet.

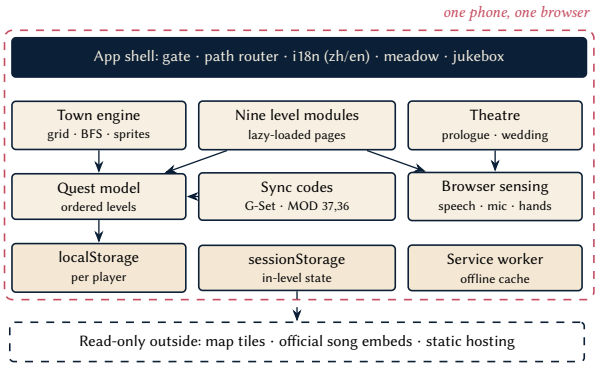


Figure 9: Architecture. Everything that holds the couple's data stays inside the browser (dashed red); the outside world is read only, and only for map tiles, song embeds and the static files themselves. A second phone joins through a code, not a server.

character follows the hybrid system ISO 7064 MOD 37,36 [13]. Writing $a_j \in \{0, \ldots, 35\}$ for the value of the j-th body character and starting from $p_0 = 36$,

$$s_j = \big((p_{j-1} + a_j) \bmod 36\big) \text{ or } 36 \text{ if } 0, \qquad p_j = 2s_j \bmod 37, \quad (3)$$

and the check character is the symbol of value $(37 - p_n) \bmod 36$. We checked the scheme exhaustively over all 1,024 valid codes. It rejects every one of the codes' single-character substitutions. Of the 2,943 transpositions of two different adjacent characters, it rejects all but five. One of those (`JI0`⟶`IJ0`) is still refused because `I` is not a player letter. The other four all swap a `0` and a `1` inside the mask (`J801`↔`J810`, `YE01`↔`YE10`): the rule above maps a running sum of 0 to 36, so a zero digit and its neighbour can trade places unseen. Substitutions and adjacent swaps are the two mistakes people make when copying a code from a chat, and in practice a swapped code that passes is still a valid code for the same player, so the union merge can at worst add levels that were not cleared. An earlier version used a plain weighted sum, which let some one-character typos through; it was replaced during a review pass, invalidating older codes. A code can also travel inside a link (`/sync?c=JVA`), which imports it when opened.

Table 4: Example progress codes: player letter, base-36 mask and the check character of Eq. 3.

Cleared	Mask m	Base 36	Code
Jingyi, levels 1–5	31	`V`	`JVA`
Jingyi, all nine	511	`E7`	`JE7Z`
Yifan, all nine	511	`E7`	`YE7O`
Yifan, none	0	`0`	`Y0C`

Three layers of storage. Cleared levels live in `localStorage` under a key per player; half-finished state inside a level (the Gomoku position reached, the stars joined so far) lives in `sessionStorage`, so a reload or a language switch inside a level does not restart it, and it is dropped when the level is cleared; and a service worker keeps the site itself. When the browser blocks storage, an in-memory store keeps progress for the visit.

5.4 The Town Engine

The town is a grid of characters (`.` grass, `#` path, `~` water, `T` tree, `O` fountain, and so on). Movement is one tile per 190 ms with an ease-out interpolation; the pure functions `step` and `faced` decide movement and facing and are unit-tested, while a class wires them to a canvas and one animation loop. A tap on the map runs a breadth-first search over walkable tiles and queues the resulting steps. The camera follows the player. The same engine takes any map, so the two cities reuse it with their own grids, roads and a basketball court.

Art as text (D5). A sprite is an array of strings, one character per pixel, and a palette that maps characters to colours. A figure is a 12×12 body per facing over four rows of legs shared between facings, so a walk cycle is the same two leg frames under different bodies, and the side view is mirrored. Tiles (grass with flowers, paths that join their neighbours, roofs in each level's colour, lit windows) are painted procedurally. The README banner, the favicon and the app icons are generated from the same strings by scripts, so a new outfit appears everywhere at once.

Daylight by city. Let h_p be the current hour in player p's time zone (`Asia/Shanghai` or `America/Chicago`). The phase is dawn for $h_p \in [6, 7)$, day for $[7, 17)$, dusk for $[17, 19)$ and night otherwise; dawn, dusk and night each lay a coloured wash over the map (night: navy at 42% opacity) and light the windows. In the two cities the person's position is $\mathrm{stopAt}(h)$ = the first stop whose hour windows contain h mod 24.

5.5 Sensing in the Browser

Three levels listen or look, and none sends a frame or a sample anywhere.

Voice. The vow and the wedding's "I do" use the Web Speech API [19] in Mandarin. All recognition alternatives are joined and normalised to Han characters and Latin letters; a vow counts when it contains 我爱你, 我愿意 and any accepted spelling of the partner's name.

Breath. Blowing out the candles reads the microphone with echo cancellation, noise suppression and gain control off, computes the root-mean-square loudness of each 1,024-sample frame, smooths it ($s \leftarrow 0.6s + 0.4\min(1, 5\,\mathrm{rms})$), and blows when $s > 0.32$. We rely on a breath being a loud, broadband rush on a phone microphone; the threshold was set by hand and has not yet been calibrated across devices.

Hands. The try-on runs MediaPipe's hand landmarker [22, 37] on the camera stream. The ring sits on the ring finger's proximal phalanx, 38% of the way from landmark 13 (the knuckle) to landmark 14; its axis follows the finger, and its "up" is the back of the hand, from the palm normal and the reported handedness. Scale comes from anatomy: the mean spacing of the middle, ring and little knuckles serves as the ruler, set to a constant of 19.5 mm (our working estimate for an adult hand, not a measured value), so

$$\mathrm{px/mm} = \tfrac{1}{2}\left(\|k_9 - k_{13}\| + \|k_{13} - k_{17}\|\right) / 19.5, \tag{4}$$

and the ring's real millimetre dimensions map to the screen. Position, rotation and scale are smoothed (factor 0.45 per frame). The shutter composes the photo with a caption, and a second card puts both players' try-on photos side by side.

Table 5: Implementation summary (version 1.17.0, 28 October 2026).

Measure	Value
TypeScript source, excluding tests (lines)	8,414
CSS (lines)	1,080
Bilingual dictionary (lines)	791
Test files / unit tests	36 / 153
Commits / versions (24–28 Oct. 2026)	109 / 28
Main bundle (min. / gzip)	114 kB / 43 kB
Precached shell (service worker)	44 files, 302 KiB
Feature folders	24
Runtime dependencies	3 (Leaflet, three.js, MediaPipe)
Servers, accounts, analytics	0

5.6 Keeping a Secret with a Test (D6)

The ring is the surprise at the end, and an interface that hints at it spoils it. Rayford, the town, the progress page and the cats all speak before the ring is reached, and their lines were written and rewritten over days. Rather than rely on vigilance, a unit test scans every key in the bilingual dictionary, except the keys shown only after the ring, and fails if any contains 戒 (ring) in Chinese, *ring* as a word in English, or any word for wedding, marriage, proposal, bride or groom in either language. One exception is listed by name: Jingyi's own joke on the Shenzhen Bay stop, which is about a flat, not a ring. The test turns a promise into a property of the build.

5.7 Implementation and Verification

Table 5 summarises the code base. The main bundle is 114 kB (43 kB compressed); the three heaviest chunks (the ring viewer with three.js, the map with Leaflet, the try-on) load only in their levels. Continuous integration runs the 153 unit tests and the strict type check on every push and deploys while publishing is switched on. Beyond unit tests, each release was checked with a scripted play-through of all nine levels in a headless browser and a sweep of every page at 320, 390 and 1,280 pixels wide in both languages. Some features cannot be verified headless. The try-on was checked with a real hand on a laptop camera (which fixed the handedness convention); speech recognition, the breath detector and the phone share sheet still await systematic testing on phones.

6 Reflections from the Design Log

Several changes the owner made to his own system show where a reasonable default failed for these two people.

From Pictionary to songs. An early level was Pictionary, after the couple's first evening of games, played on one device by turns. The owner replaced it with *Our Songs*. Our reading of the change is that a drawing game played by turns assumes two people at one screen, which this couple never are, whereas a song can be recognised alone and still brings the other person's music with it.

From reveal to photograph. The ring level was at first cleared by watching the reveal. The owner changed the rule so that only the try-on photo clears it: the level asks for an act, not for attention, and leaves an image to send.

Getting the person right. Generated defaults guessed that Yifan takes the bus and that Jingyi takes the metro, and named the baker in the prologue after Yifan's nickname. Each was corrected: Yifan drives; Jingyi drives a Tesla; the baker is Yifan himself. For an audience of one, a plausible guess about her life reads as not knowing her.

One question at the door. The gate's placeholder text once spelled out the answer. A review pass removed it. The gate protects little (Section 7.2), but a door that answers its own question is no longer a question only the two of them can answer.

7 Discussion

7.1 Memory as Mechanics

Hassenzahl et al. found *memories* to be one strategy among six [11]; in Yi Town it is the grammar of the whole system. The interesting design work was not choosing which memories to include but finding the *rule* each memory implies. A first game on a whiteboard implies Gomoku, but "win-in-one" implies something more: that each step is a small discovery, and that ten discoveries can add up to a shape. The heart then recurs: as gold stones on the board, as strawberries on the cake, as the outline the player follows. Repeated motifs let a small system feel authored rather than assembled.

7.2 Privacy as Intimacy, Not Security

The gate is not security. Its answers are dates, and a determined attacker could enumerate them in moments; the SHA-256 hash only keeps the answer from being read in the source. Its purpose is social: the site asks a question only the two of them can answer, and that question is itself a small ritual of recognition. Likewise, keeping all progress on the device (D3) is less about threat models than about where the couple's things live. A voice note in a mailbox in a town on her phone is a different object from the same audio on a server. Local-first software [18] usually argues for ownership and longevity; here the argument is closeness.

7.3 Thirteen Hours

Asynchrony is often treated as a problem to be bridged with presence. Yi Town treats it as material. Each player's town is lit by their own clock, so the two never see the same sky; the partner by the fountain is a message left, not a presence; and the birthday prologue is staged precisely on the gap, a morning in Texas that is the last hour before midnight in Shenzhen. The prologue's line "In Yi Town there is no thirteen-hour time difference" works because everything before it has shown the difference. This echoes slow technology's interest in time as a design material [10, 28].

7.4 Secrecy as a Design Material

Surprise is fragile in software: any string, any loading screen, any helpful guide can leak it. We found it useful to treat the surprise as an invariant and to test it, as one tests that a list is sorted. The secrecy test also shaped the writing: Rayford had to explain why he wears a suit without saying why, and the answer is paid off only in the wedding. Ambiguity, here, was not a resource [8] so much as a promise held open.

7.5 Two Languages, One Voice

Writing every string twice, by the same author, made the two languages talk to each other. Puns cross the boundary: the title 仪见钟情 replaces the 一 of 一见钟情 (love at first sight) with the 仪 of Jingyi's name, a pun that cannot be translated, so the English site names it and explains it. Nine candles mean "for a long time" only in Chinese, and the English caption says so. Parity (D4) meant that neither reader is given the translated version.

7.6 Building with an AI Pair Programmer

Yi Town was built in five days by one person working with an AI coding assistant. The assistant wrote most of the code from the owner's descriptions; the owner chose every memory, line, rule and correction. Two practices kept the result personal rather than generic: the owner corrected defaults about their lives as soon as he saw them (Section 6), and invariants that mattered to him were written as tests rather than as instructions, which an assistant can forget.

7.7 Limitations and Future Work

Yi Town is a sample of two, and its first real play is Jingyi's birthday; this paper describes the artefact and its making, not her experience, which we will report after it has happened, in her own words. The two players cannot yet play at the same time over a network; the songs depend on official embeds that may not play in every region; the quizzes can be brute-forced by guessing; several features (speech, breath, camera) depend on browser support that varies across phones and have not been tested systematically on them; and every number in this paper describes the system, not its use. The owner's next steps are to rewrite the two letters in his own words, add voice notes, and build a second map, a long road from Shenzhen to College Station.

8 Conclusion

We described Yi Town, a game for two people who live thirteen hours apart, built by one of them out of what they share. Its architecture is deliberately small: a grid, some strings that are pictures, a grow-only set, a four-character code, a test that keeps a secret. Its content is deliberately large: a vow, six places, a tattoo, forty days, a heart of stones, six songs, two constellations, five poems, a ring, a cake and a wedding. For an audience of exactly one other person, technical choices and emotional ones turned out to be the same choices, and we hope the commitments listed here are useful to others who build for the person on the other side of the ocean.

Code and Data Availability

The complete source of Yi Town, with its tests, design notes, changelog and the Blender script for the rings, is available at https://github.com/rayford295/yi-town; this paper describes version 1.17.0 (28 October 2026). The repository also holds the personal photographs, voice notes and letters that the game is made of, published with the consent of both authors. The street-view photographs come from Wikimedia Commons under the Creative Commons licences credited in the game, and the *Wedding March* recording is in the public domain. The paper's source and figures are in the repository's `paper/` folder.

Declaration of Generative AI Use

The authors used Claude (Anthropic), through the Claude Code assistant, in two ways. First, for the software: the assistant wrote most of the source code of Yi Town from the authors' descriptions, while the authors chose every memory, line, rule and correction and reviewed and tested the system. Second, for this manuscript: the assistant was used to draft and polish the text from the repository and the authors' notes, to prepare the figures, and to check the references against bibliographic databases. The authors reviewed and edited all text, verified every reference and every number reported, and take full responsibility for the content. No AI system is an author of this work.

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